

fractions, with 1 being the time 24:00 or 00:00. 18:00 has true value of 0.75 because it is three quarters of 24 hours.

For more details, see 'How Microsoft Excel stores dates and times' in Excel 'Help'.

To see the true value of a date and/or time simply format the cell as 'General'. For example, the date and time '3/July/2002 3:00:00 PM' has a true value of 37440.625 with the number after the decimal representing the time and the 37440 being the serial value for 3/July/2002.

3.4.1.5 Adding Past 24 Hours

We can simply add times by using the Sum function or a simple addition sign, e.g., =Sum(A1:A5) would result in the total hours if A1:A5 contained valid times. There is however a big 'Gotcha!' and that is, unless told otherwise, Excel will not add past 24 hours. This is because when the time value exceeds 24 hours (true value of 1) it rolls into a new day and starts again. To force Excel not to default back to a new day after 24 hours we use a cell format of 37:30:55 or a 'Custom' format of [h]:mm:ss.

The same format can be used to get the total minutes or seconds of a time, for example to get the total minutes of the time 24:00 format the cell as [m] and you will get 1440 (see why below). To get the total seconds use a custom format of [s] and you get 86400.

3.4.1.6 Time And Date Calculations

If we wish to use these real time values in other calculations there are a few 'magic' numbers to keep in mind.

1. 60 (sixty minutes or sixty seconds)
2. 3600 (60 seconds * 60 minutes)
3. 24 (24 hours)
4. 1440 (60 minutes * 24 hours)
5. 86400 (24 hours * 60 minutes * 60 seconds)